DO NOW, AO1 REVISION

Density of materials

Particle model of matter
Paper 1, Combined and Separate
13 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Define density.**	The mass per unit volume of a substance.	☐
2	**State the standard unit of volume.**	Cubic metre (m^3)	☐
3 ★	**State the standard unit of density.**	Kilogram per cubic metre (kg/m^3)	☐
4	**Which state of matter can be compressed easily?**	Gas	☐
5	**Explain why solids and liquids cannot be compressed easily.**	Their particles are already touching.	☐
6 ★	**Describe the motion of particles in a solid.**	They vibrate around a fixed position.	☐
7	**Describe the motion of particles in a liquid.**	They move randomly, sliding past one another.	☐
8	**Describe the motion of particles in a gas.**	They move quickly in random directions.	☐
9	**State the equation linking density, mass and volume.**	density = mass ÷ volume ($\rho = m \div V$)	☐
10	**Describe the arrangement of the particles in a solid.**	Very close together in a regular pattern.	☐
11	**Describe the arrangement of the particles in a liquid.**	Close together, in a random arrangement.	☐
12	**Describe the arrangement of the particles in a gas.**	Far apart, in a random arrangement.	☐
13 ★	**Explain why gases have a much lower density than solids and liquids.**	The particles are far apart, so there is less mass in a given volume.	☐

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